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Information Transduction With Multiple States

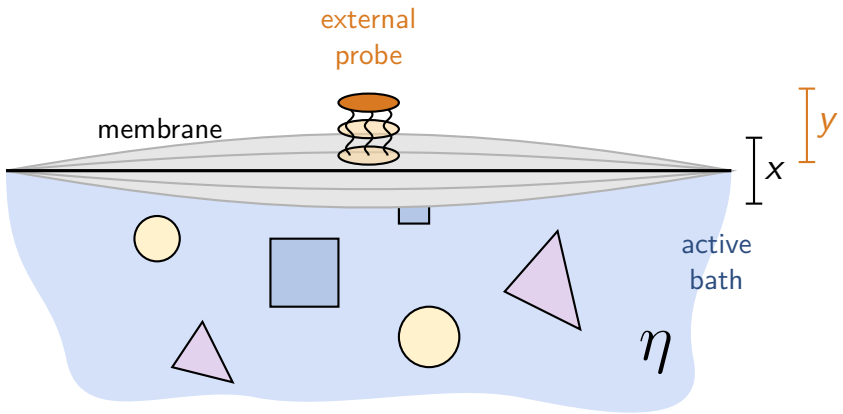
Master's Degree in Physics of Data

William Conte
Internal supervisor: Prof. Sandro Azaele
External supervisor: Dr. Daniel Maria Busiello

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Transduction

A system captures information about an unobservable external world and adapts its internal dynamics accordingly (cells, metabolic pathways, chemical networks).



- **Active Bath (η):** inaccessible degrees of freedom driving the system (e.g., cytoskeleton).
- **Membrane (x):** intermediate variable relaying information (e.g., lipid bilayer).
- **Probe (y):** the only variable accessible to an observer (e.g., optical tweezers).

The Transduction Problem



*"How can a system extract the maximum amount of information from an **inaccessible active bath**, while spending the **minimum possible energy**?"*

The Transduction Problem

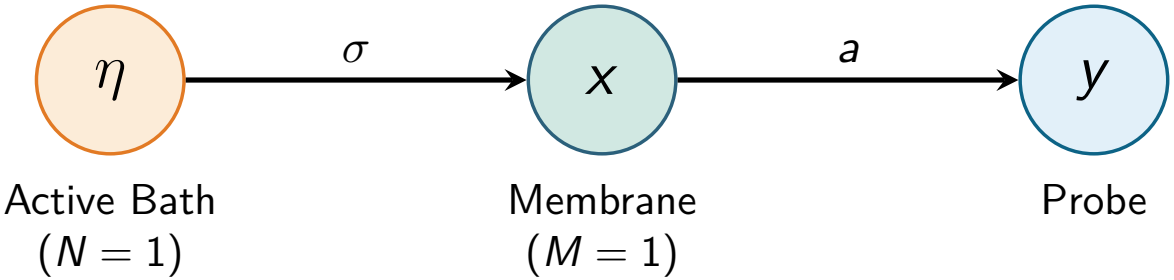
*"How can a system extract the maximum amount of information from an **inaccessible active bath**, while spending the **minimum possible energy**?"*

1. The Foundation
Three-State Model

→

2. Generalization
Complex Topologies

The Three-State Model



Langevin Equations

$$\tau_\eta \dot{\eta} = -\eta + \sqrt{2D_\eta \tau_\eta}\,\xi_\eta$$
$$\tau_x \dot{x} = -x + \sigma\eta + \sqrt{2D_x \tau_x}\,\xi_x$$
$$\tau_y \dot{y} = -y + ax + \sqrt{2D_y \tau_y}\,\xi_y$$

Vectorial Form

$$\dot{\boldsymbol{z}} = -\boldsymbol{A}\boldsymbol{z} + \tilde{\boldsymbol{D}}\boldsymbol{\xi}(t) \quad \text{with } \boldsymbol{z} = \begin{pmatrix} y \\ x \\ \eta \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} 1/\tau_y & -a/\tau_y & 0 \\ 0 & 1/\tau_x & -\sigma/\tau_x \\ 0 & 0 & 1/\tau_\eta \end{pmatrix}$$

$$\tilde{\boldsymbol{D}} = \sqrt{2}\begin{pmatrix} \sqrt{D_y/\tau_y} & 0 & 0 \\ 0 & \sqrt{D_x/\tau_x} & 0 \\ 0 & 0 & \sqrt{D_\eta/\tau_\eta} \end{pmatrix}$$

Information vs Dissipation

Information

MI for a gaussian stationary state:

$$I(x;y) = \frac{1}{2} \ln \left[\frac{\Sigma_{xx}\Sigma_{yy}}{\Sigma_{xx}\Sigma_{yy} - \Sigma_{xy}^2} \right]$$

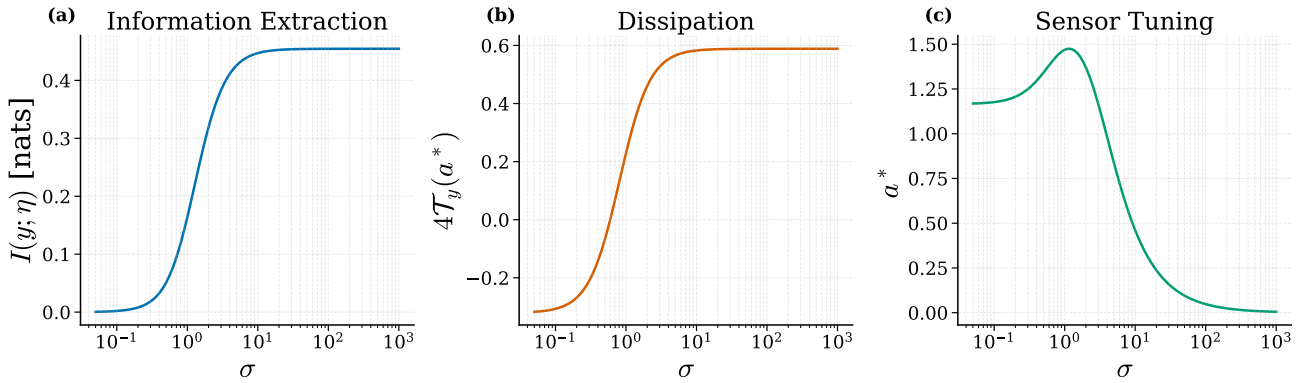
Energy Expenditure

Relative entropy rate $\dot{S}^{rel} = \dot{S}_a - \dot{S}_{a=0} \Rightarrow$ probe traffic:

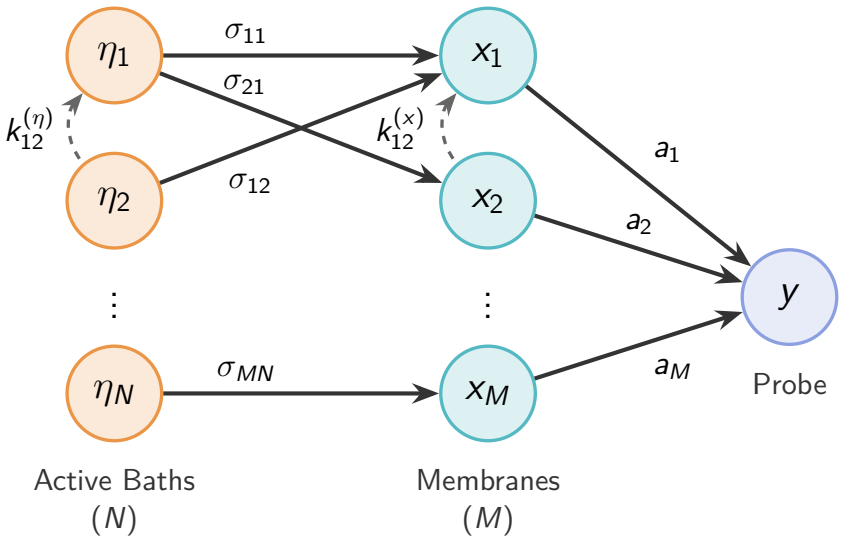
$$4\mathcal{T}_y(a) = [\boldsymbol{D}^{-1}\boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}^T - 2\boldsymbol{A}]_{yy}$$

Pareto Optimization Functional

$$\mathcal{L}(a) = \lambda I(x;y) - (1-\lambda)4\mathcal{T}_y(a), \qquad a^* = \arg\max_a \mathcal{L}(a)$$



Multiple state model



$$\boldsymbol{z} = (y, \boldsymbol{x}^T, \boldsymbol{\eta}^T)^T, \quad \dot{\boldsymbol{z}} = -\boldsymbol{A}\boldsymbol{z} + \tilde{\boldsymbol{D}}\boldsymbol{\xi}(t):$$

$$\boldsymbol{A} = \begin{pmatrix} \frac{1}{\tau_y} & -\frac{1}{\tau_y}\boldsymbol{a}^T & \boldsymbol{O} \\ \boldsymbol{O} & \boldsymbol{T}_x^{-1}\boldsymbol{A}_x & -\boldsymbol{T}_x^{-1}\boldsymbol{\sigma} \\ \boldsymbol{O} & \boldsymbol{O} & \boldsymbol{T}_\eta^{-1}\boldsymbol{A}_\eta \end{pmatrix}$$

- $\boldsymbol{A}_x, \boldsymbol{A}_\eta$: self-decays and structural couplings k; $\boldsymbol{T}_x, \boldsymbol{T}_\eta$: diagonal relaxation times; $\tilde{\boldsymbol{D}}$ diagonal.

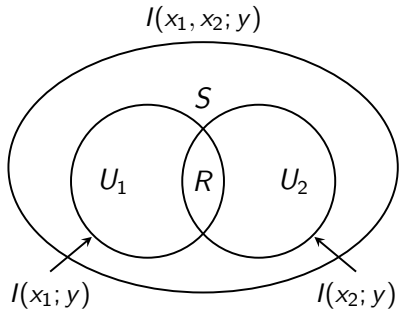
$k_{ij}^{(\dots)}$ acts as an internal structural coupling $\rightarrow$ needed to model inhibition ($k < 0$) or co-activation ($k > 0$).

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PID Equation

$$I(x_1, x_2; y) = U(x_1 \setminus x_2; y) + U(x_2 \setminus x_1; y) + R(x_1, x_2; y) + S(x_1, x_2; y)$$

- **Redundancy (R):** overlapping info available from either source alone.
- **Unique Info (U):** info provided exclusively by one source.
- **Synergy (S):** emergent info accessible only from the joint state.



Gaussian evaluation, MMI closure and Whole-Minus-Sum

$$I(\boldsymbol{x}; y) = \frac{1}{2} \ln \left[\frac{\Sigma_{yy} \det(\boldsymbol{\Sigma}_{xx})}{\det(\boldsymbol{\Sigma}_{joint})} \right], \quad R = \min\{I(x_1; y), I(x_2; y)\}$$

$$\text{WMS} = I(x_1, x_2; y) - \Big(I(x_1; y) + I(x_2; y)\Big) = S - R$$

Multi-Source Optimization Scheme ($M = 2$)



Parameterized Pareto Functional

$$\mathcal{L}(a_1, a_2; \lambda, \gamma) = \lambda \Big[U(x_1 \setminus x_2; y) + U(x_2 \setminus x_1; y) + 2\gamma S(x_1, x_2; y) + 2(1-\gamma) R(x_1, x_2; y) \Big] - (1-\lambda) 4 \mathcal{T}_y(a_1, a_2)$$

The parameter $\gamma \in [0, 1]$ dictates the information-harvesting strategy:

- $\gamma = 1 \implies$ **Synergy-driven:** prioritizes emergent information.
- $\gamma = 0.5 \implies$ **Balanced:** recovers standard joint MI $I(x_1, x_2; y)$.
- $\gamma = 0 \implies$ **Redundancy-driven:** favors shared, robust signals.

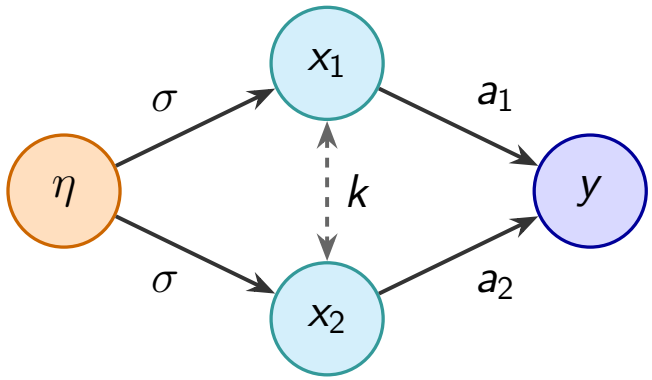
Fair comparison: architectures are compared at **equal dissipation**, $\dot{S}^{rel}_{M,N} = \dot{S}^{rel}_{M=1,N=1}(a^*)$, which restricts the search to an $(M-1)$-dimensional manifold.

Topology $M = 2$, $N = 1$

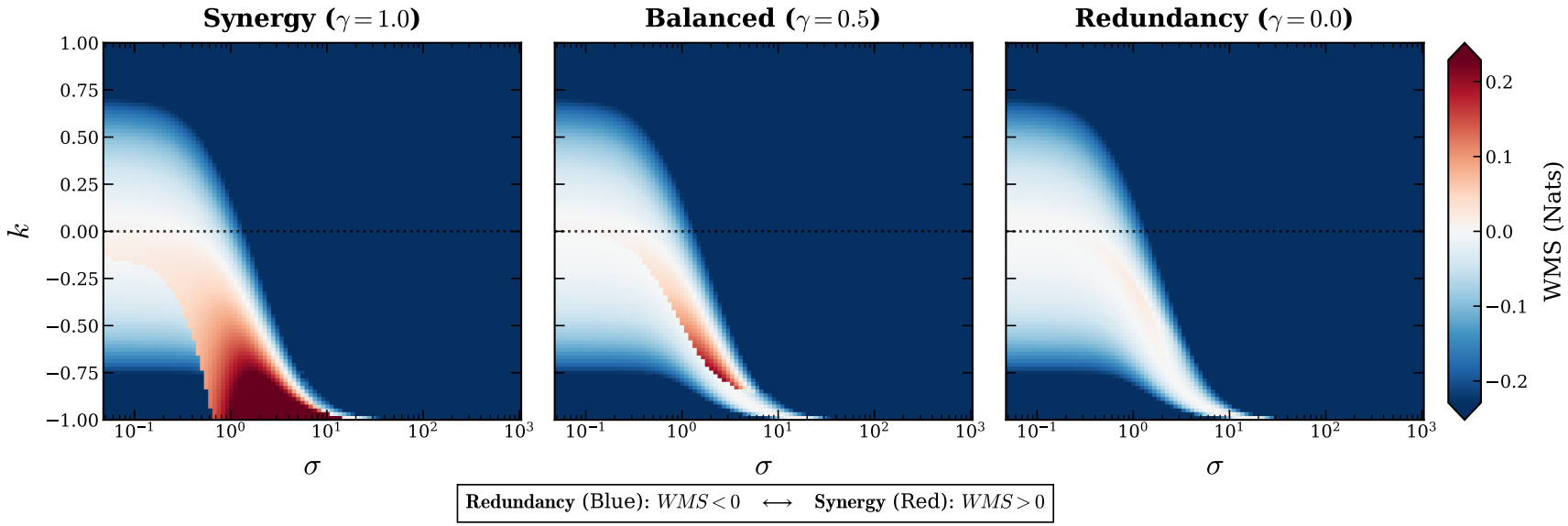
Both membrane states receive the same drive σ and interact through k:

$$\dot{x}_1 = -\frac{x_1}{\tau_{x_1}} + \frac{k}{\tau_{x_1}} x_2 + \frac{\sigma}{\tau_{x_1}} \eta + \sqrt{\frac{2D_{x_1}}{\tau_{x_1}}} \xi_{x_1}(t)$$

- x_2 obeys the symmetric equation $(1 \leftrightarrow 2)$; the probe reads $a_1 x_1 + a_2 x_2$.
- Stable iff $|k| < 1$ (eigenvalues $(1 \pm k)/\tau_x$).

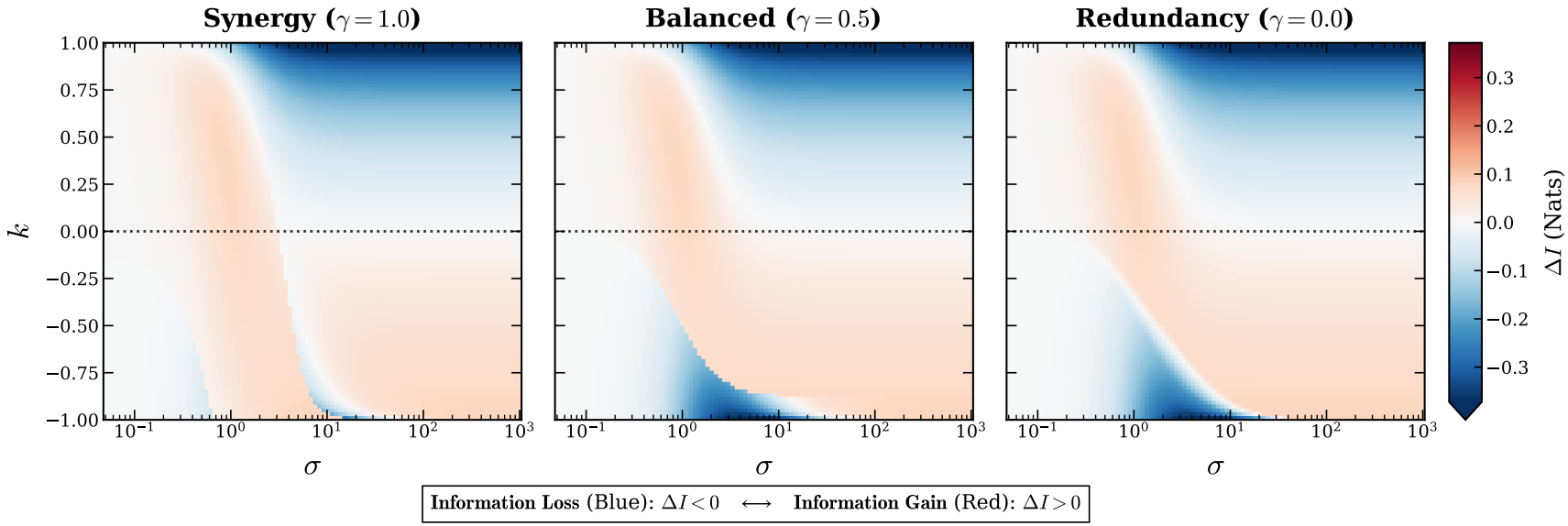


Whole Minus Sum (WMS) — $M = 2,\ N = 1$



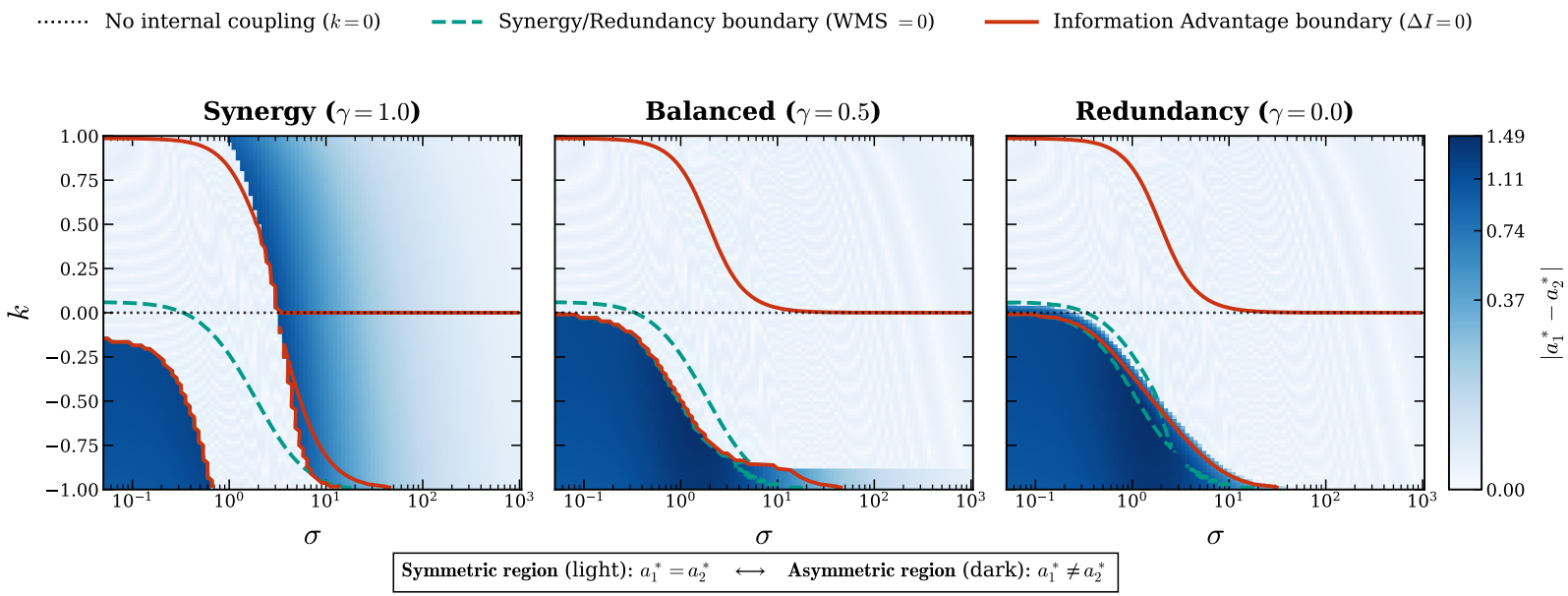
- $k > 0 \rightarrow$ Structural co-activation drives a purely redundant regime.
- $k < 0 \rightarrow$ Redundancy generally prevails, but synergy emerges at intermediate σ values.
- Synergy emerges in this intermediate range because the environmental drive is strong enough to be captured, but the internal structural inhibition prevents simple redundant extraction.

$\Delta I(y;\eta)$ — $M = 2,\ N = 1$



- $k > 0 \rightarrow$ info gain at low-to-mid σ. Performance drops at high σ due to slower relaxation $(1 - k)/\tau_x$ and noise accumulation.
- $k < 0 \rightarrow$ faster relaxation improves fluctuation tracking. Guarantees information gain at higher σ across all strategies.
- Synergy is the most robust strategy for $k < 0$.

Coupling Asymmetry $|a_1^* - a_2^*|$ — $M = 2,\ N = 1$



(WMS and ΔI maps: see the previous slides)

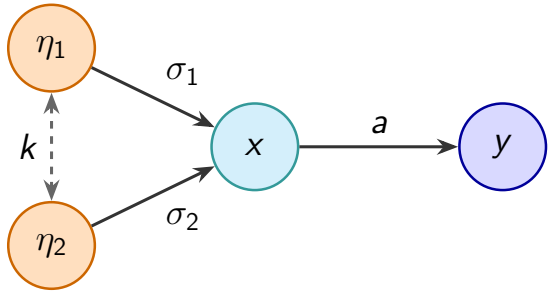
- $k > 0 \rightarrow$ in the synergy strategy, transitioning to a single active channel perfectly matches the loss of information advantage.
- $k < 0 \rightarrow$ restoring structural symmetry coincides perfectly with information gain ($\Delta I > 0$) and synergy onset.

Topology $M = 1$, $N = 2$

The membrane integrates two baths, coupled by k:

$$\dot{x} = -\frac{x}{\tau_x} + \frac{\sigma_1}{\tau_x}\eta_1 + \frac{\sigma_2}{\tau_x}\eta_2 + \sqrt{\frac{2D_x}{\tau_x}}\xi_x(t)$$

$$\dot{\eta}_1 = -\frac{\eta_1}{\tau_\eta} + \frac{k}{\tau_\eta}\eta_2 + \sqrt{\frac{2D_{\eta_1}}{\tau_\eta}}\xi_{\eta_1}(t)$$



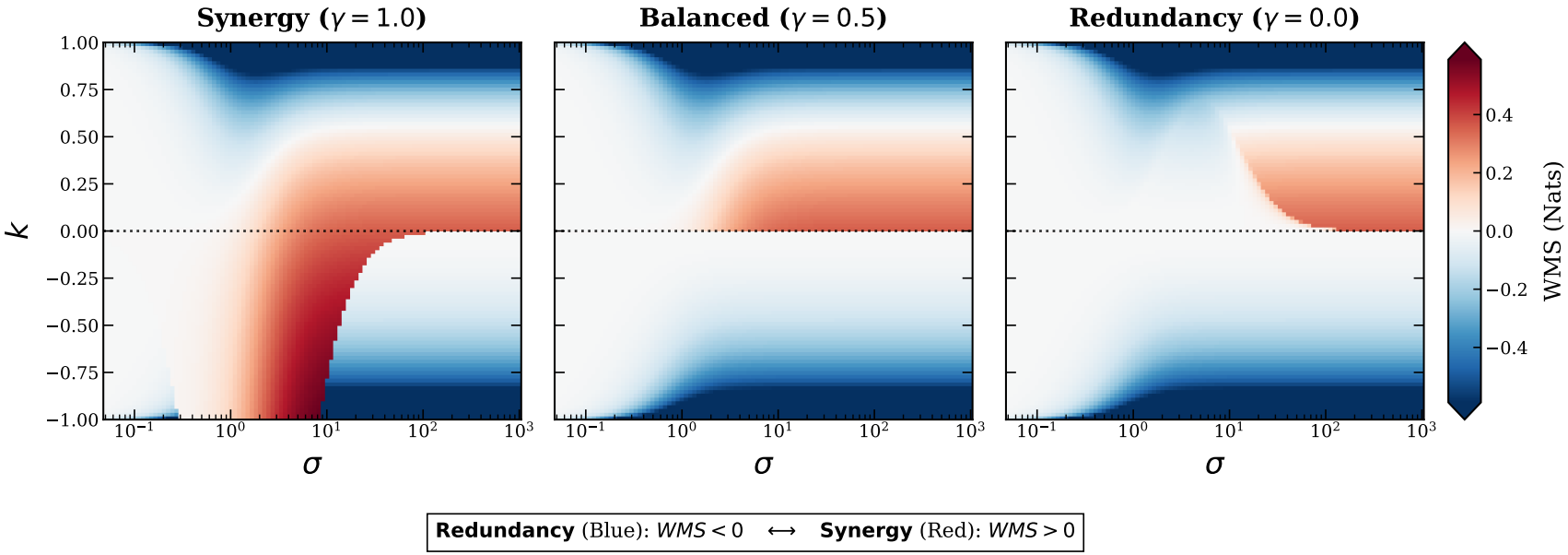
η_2 symmetric; stable iff $|k| < 1$

Change of agency: membrane-centric optimization

The optimized couplings are now the environmental drives σ_1, σ_2: the functional involves the PID of $(\eta_1, \eta_2; x)$ and the **membrane traffic** $4\mathcal{T}_x$, at equal dissipation.

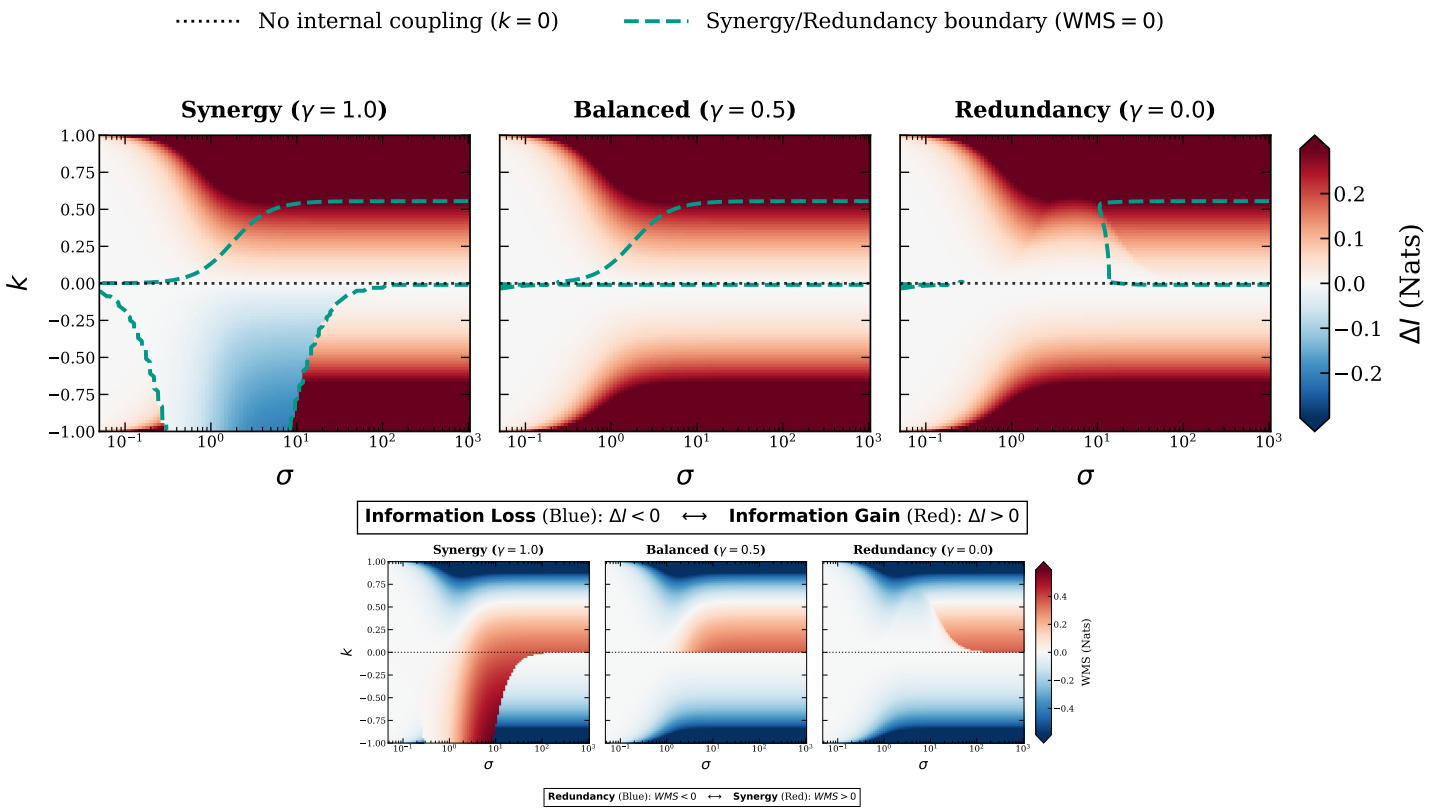
$$\mathcal{L}(\sigma_1, \sigma_2; \lambda, \gamma) = \lambda\Big[U(\eta_1 \setminus \eta_2; x) + U(\eta_2 \setminus \eta_1; x) + 2\gamma S(\eta_1, \eta_2; x) + 2(1-\gamma)R(\eta_1, \eta_2; x)\Big] - (1-\lambda)4\mathcal{T}_x(\sigma_1, \sigma_2)$$

Whole Minus Sum (WMS) — $M = 1,\ N = 2$



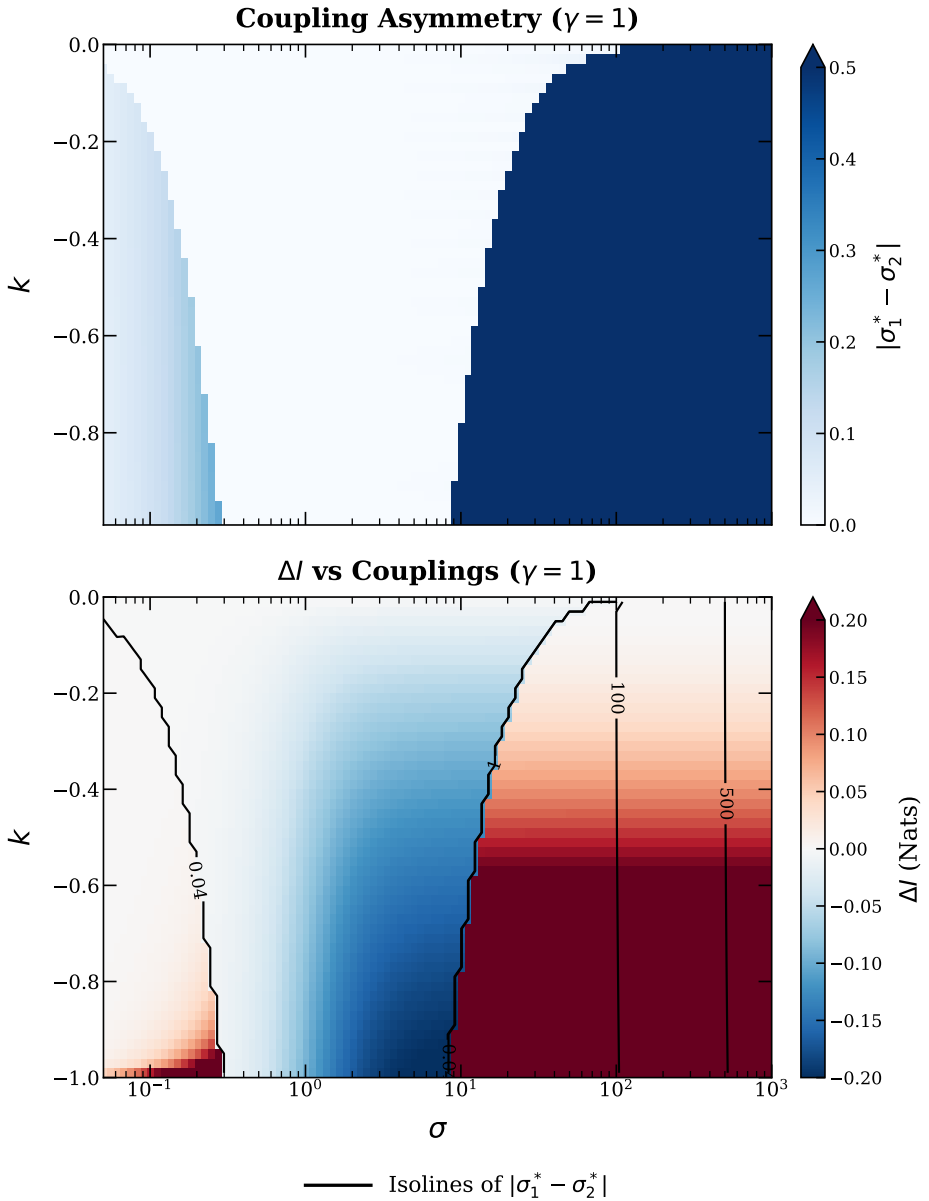
- $k > 0 \rightarrow$ synergy emerges only for weak couplings ($k \lesssim 0.5$). Higher structural correlation traps the system in redundancy.
- $k < 0 \rightarrow$ under the synergy-driven strategy, strong anti-correlation ($k \rightarrow -1$) restricts the synergistic region to lower σ values due to opposed fluctuations.

$\Delta I(y;\eta)$ vs WMS — $M = 1$, $N = 2$



- **The Redundancy Principle:** Across all coupling regimes, redundancy is structurally tied to a net information gain ($\Delta I > 0$).
- **Performance vs. Synergy ($k < 0$):** WMS boundaries perfectly match performance: synergy $\implies$ information loss ($\Delta I < 0$).

$\Delta I(y;\eta)$ vs $|\sigma_1^* - \sigma_2^*|$



Synergy-driven strategy ($\gamma = 1$), $k < 0$

Synergy

$\Delta I < 0$

Symmetry

$\sigma_1^* = \sigma_2^*$

Conclusions



- Adding a single state to the membrane ($M = 2$) or to the active bath ($N = 2$) already produces non-trivial information-extraction patterns, at **equal dissipation**: any ΔI is due to the architecture alone.
- **The optimal strategy follows the topology.**
  Many sensors on few environments $\Rightarrow$ **synergy**
  Many environments with few sensors $\Rightarrow$ **redundancy**.
- **Synergy requires all couplings active**, while redundancy is more flexible and can also emerge when one pathway is switched off. Hence no universal optimal architecture: sensors should adapt to the statistics of the environment.
- **Limits and outlook:** linear–Gaussian setting with MMI-based PID; next steps are non-linear couplings, feedback from the probe, force-free validation with the VSR, and adaptive couplings under abrupt environmental changes.

Thank you for the attention



Backup slides

Topology $M = 2$, $N = 1$: Full Equations

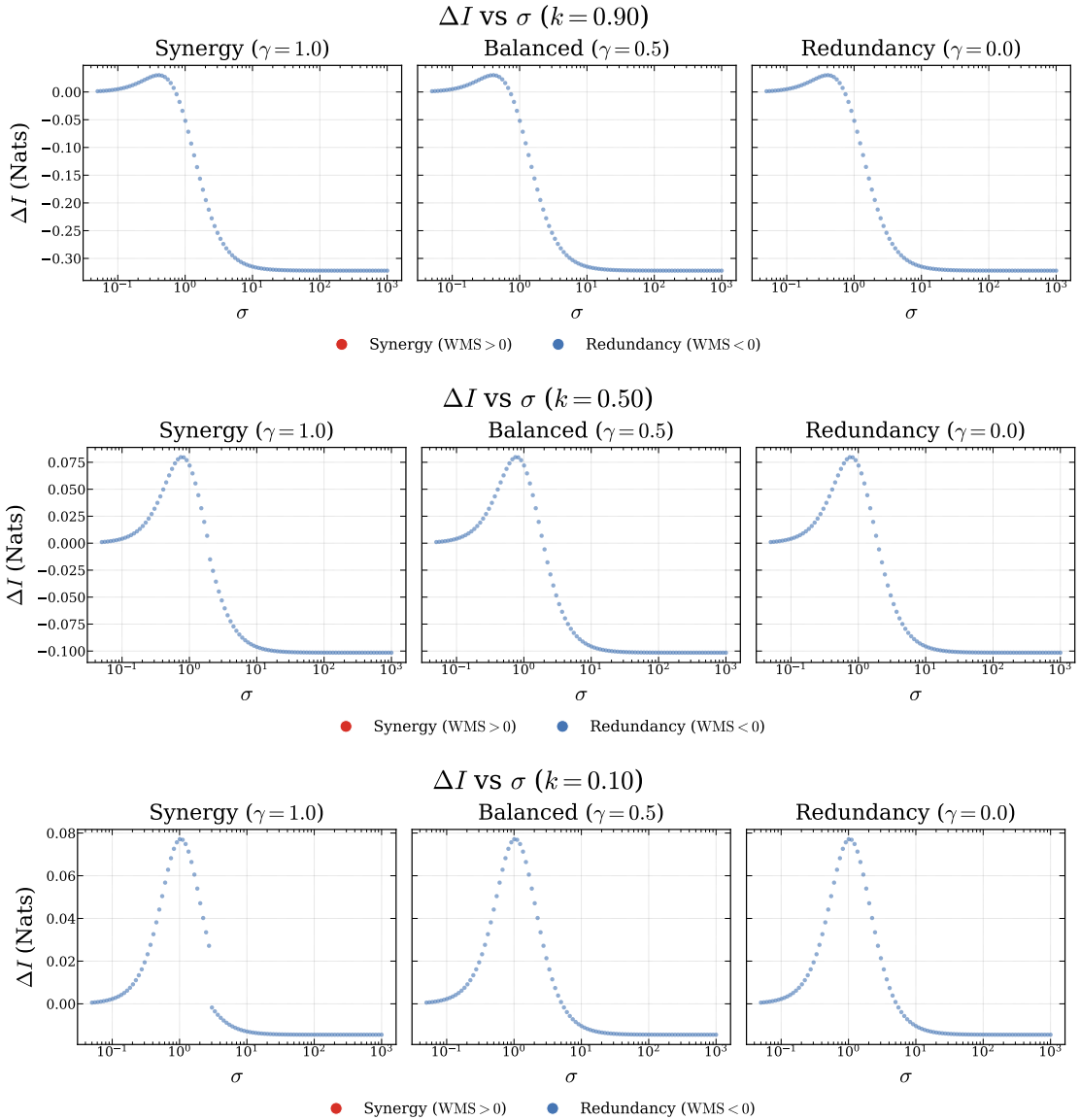
$$\begin{cases} \dot{y} = -\frac{y}{\tau_y} + \frac{a_1}{\tau_y} x_1 + \frac{a_2}{\tau_y} x_2 + \sqrt{\frac{2D_y}{\tau_y}} \xi_y(t) \\ \dot{x}_1 = -\frac{x_1}{\tau_{x_1}} + \frac{k}{\tau_{x_1}} x_2 + \frac{\sigma}{\tau_{x_1}} \eta + \sqrt{\frac{2D_{x_1}}{\tau_{x_1}}} \xi_{x_1}(t) \\ \dot{x}_2 = -\frac{x_2}{\tau_{x_2}} + \frac{k}{\tau_{x_2}} x_1 + \frac{\sigma}{\tau_{x_2}} \eta + \sqrt{\frac{2D_{x_2}}{\tau_{x_2}}} \xi_{x_2}(t) \\ \dot{\eta} = -\frac{\eta}{\tau_\eta} + \sqrt{\frac{2D_\eta}{\tau_\eta}} \xi_\eta(t) \end{cases}$$

Topology $M = 1$, $N = 2$: Full Equations

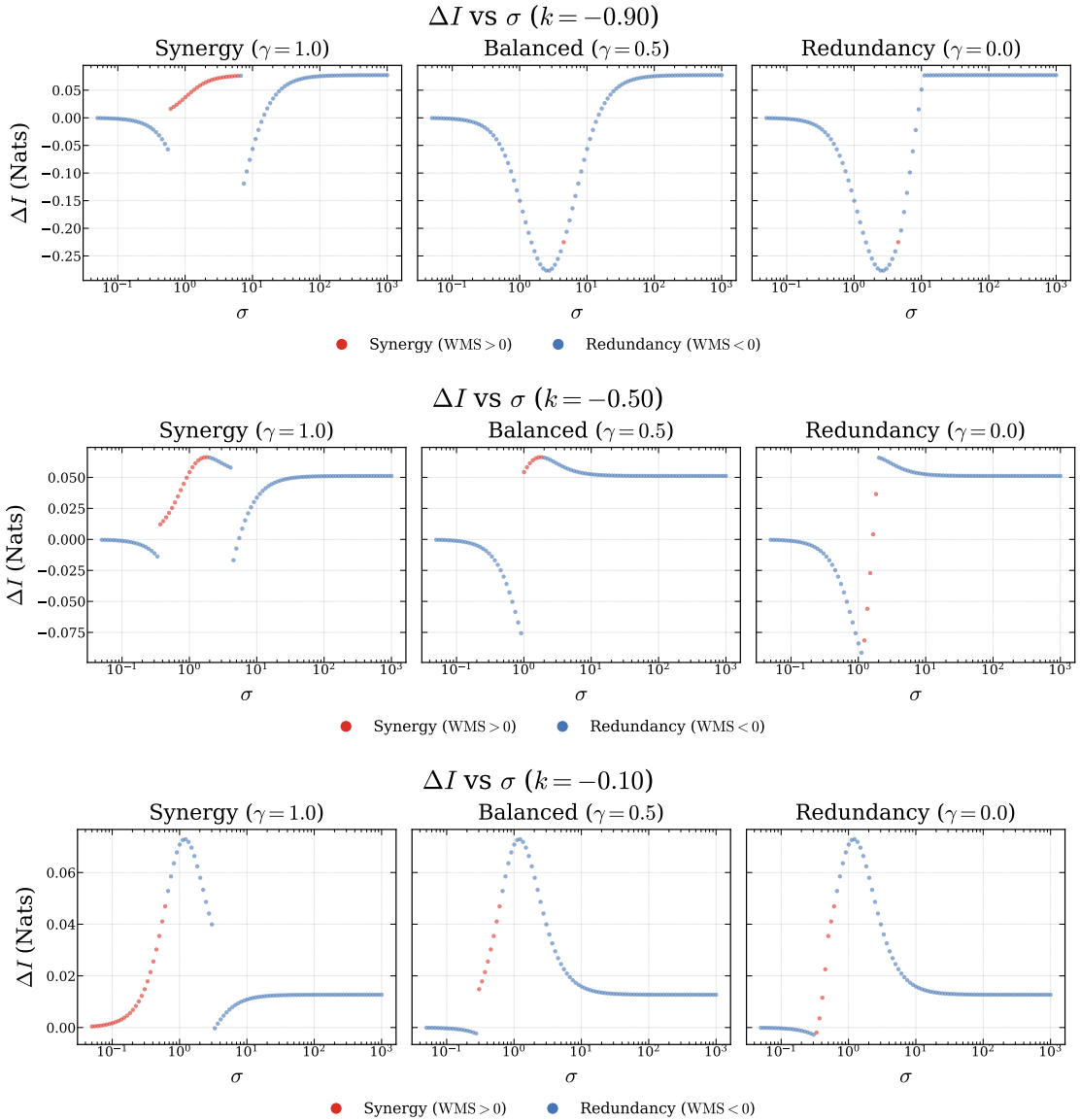


$$\begin{cases} \dot{y} = -\frac{y}{\tau_y} + \frac{a}{\tau_y} x + \sqrt{\frac{2D_y}{\tau_y}} \xi_y(t) \\ \dot{x} = -\frac{x}{\tau_x} + \frac{\sigma_1}{\tau_x} \eta_1 + \frac{\sigma_2}{\tau_x} \eta_2 + \sqrt{\frac{2D_x}{\tau_x}} \xi_x(t) \\ \dot{\eta}_1 = -\frac{\eta_1}{\tau_\eta} + \frac{k}{\tau_\eta} \eta_2 + \sqrt{\frac{2D_{\eta_1}}{\tau_\eta}} \xi_{\eta_1}(t) \\ \dot{\eta}_2 = -\frac{\eta_2}{\tau_\eta} + \frac{k}{\tau_\eta} \eta_1 + \sqrt{\frac{2D_{\eta_2}}{\tau_\eta}} \xi_{\eta_2}(t) \end{cases}$$

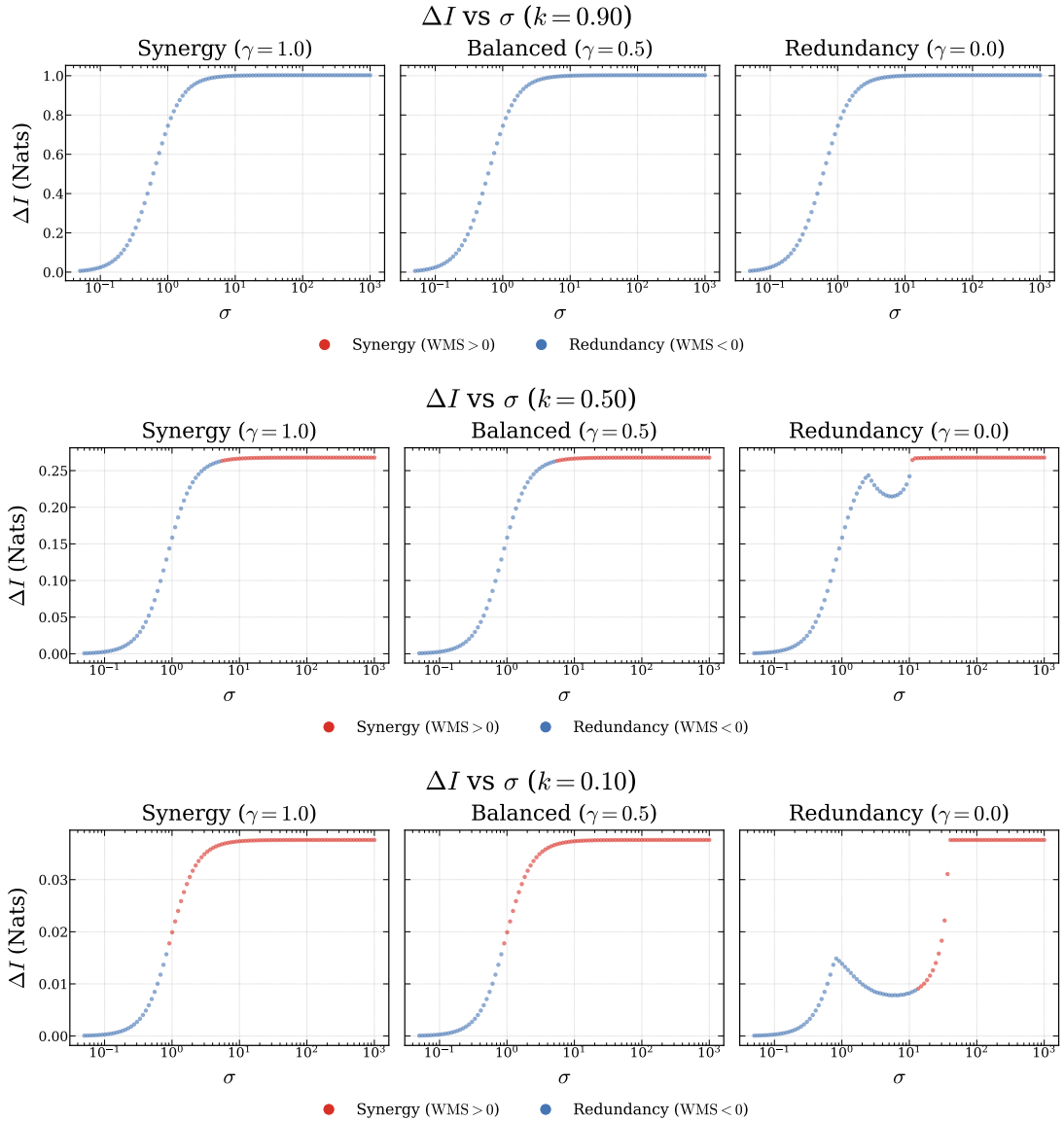
ΔI vs WMS - M=2, N=1 (1)



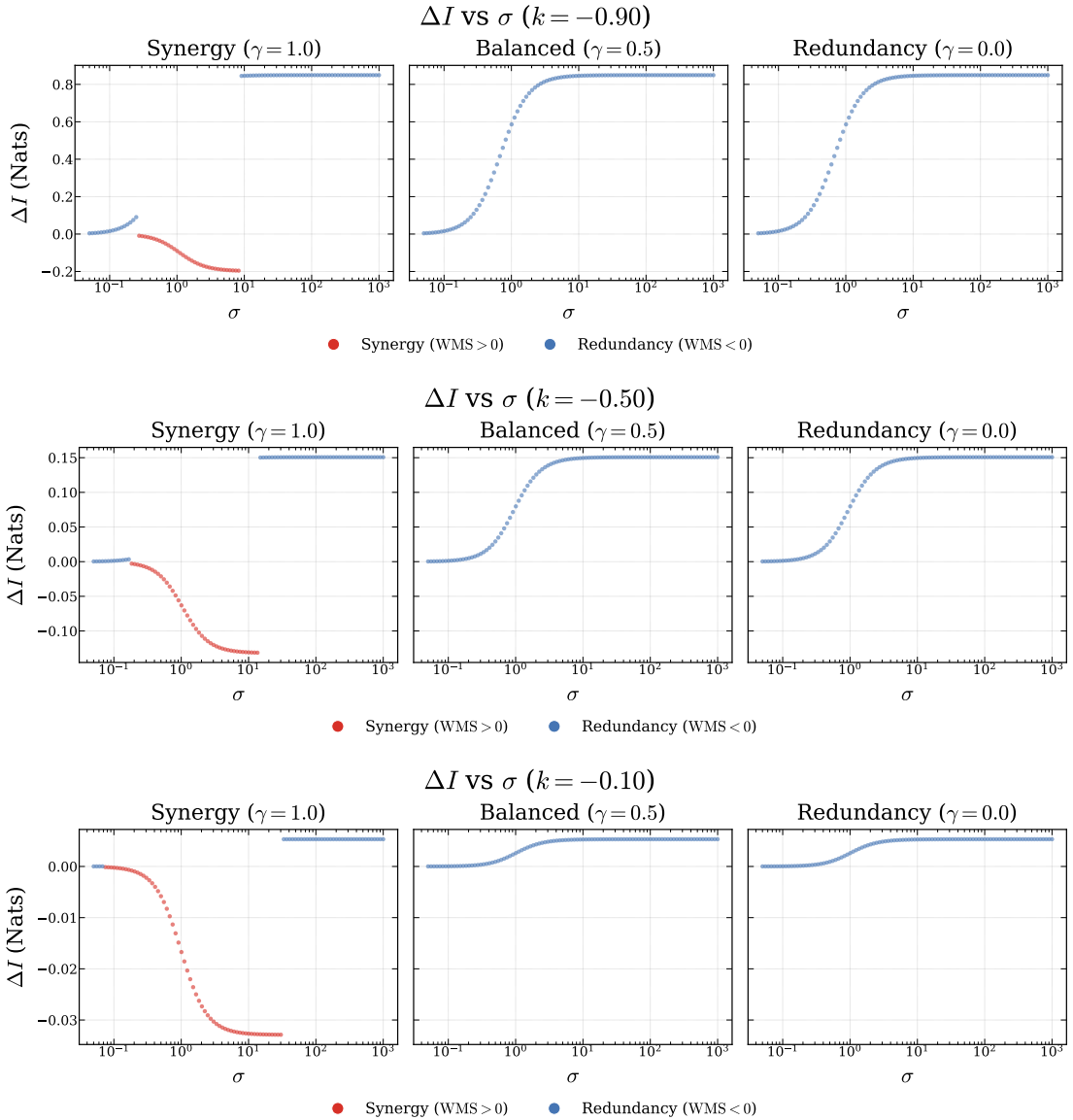
ΔI vs WMS - M=2, N=1 (2)



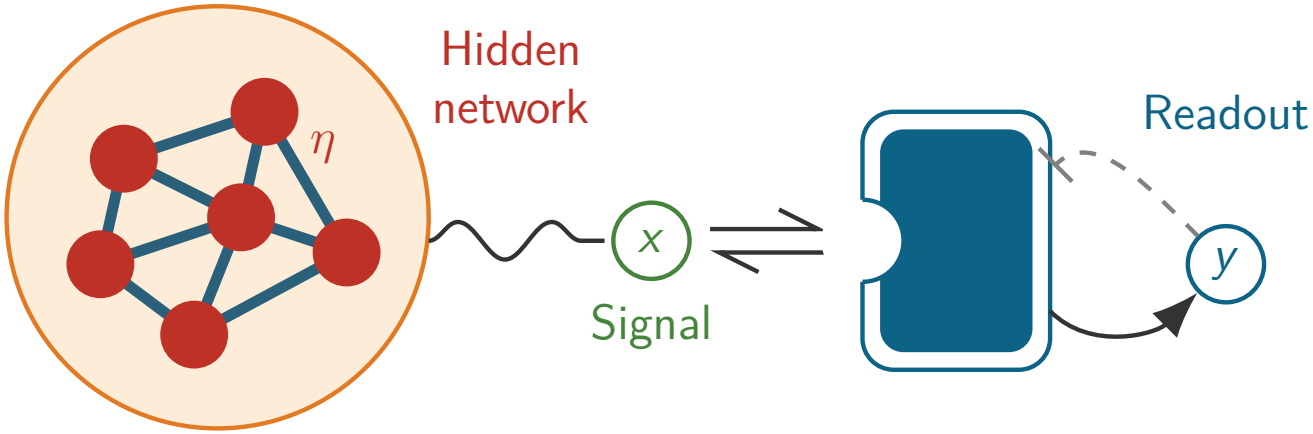
ΔI vs WMS - M=1, N=2 (1)



ΔI vs WMS - M=1, N=2 (2)



Example: Chemical Network



Chemical Transduction Network: Reaction Scheme

A hidden chemical network H stimulates the production of a signaling molecule X, which binds a receptor to produce an internal readout Y:

$$\sum_j \alpha_j H_j \rightleftharpoons \sum_j \beta_j H_j$$

$$\sum_j \gamma_j H_j + E \rightleftharpoons C_E \to E + X$$

$$X + R \rightleftharpoons C_R \to R + Y, \qquad Y \to \emptyset$$

- $\alpha_j, \beta_j, \gamma_j$: generic stoichiometric coefficients, so the internal dynamics of the hidden species H_j stays fully general.
- The first reaction is the internal dynamics of the hidden network; the second says that the enzyme E produces X from a combination of hidden species (via the complex C_E).
- Kramers-Moyal expansion of the chemical master equation, truncated at second order (large copy numbers) $\Rightarrow$ chemical Langevin equation: mass-action drift and multiplicative noise, non-diagonal, coupling H, X, C_E, C_R, Y.

Langevin Description and Timescale Separation

General signaling model for the concentrations:

$$\dot{H}_i = A_i^{(H)}(H) + b_i^{(H)}(H) \cdot \xi_H$$
$$\dot{X} = A^{(X)}(H, X, Y) + b^{(X)}(H, X, Y)\, \xi_X$$
$$\dot{Y} = A^{(Y)}(X, Y) + b^{(Y)}(X, Y)\, \xi_Y$$

- H_i: hidden species, interacting through a nonlinear force field and through mixed white noises; X, Y: the species accessible to the sensor.
- Conservation laws: $R = R_{tot} - C_R$, $E = E_{tot} - C_E$.
- The complexes C_R, C_E are **fast variables**, and all couplings between H, X, Y go through them $\Rightarrow$ integrate them out of the Fokker–Planck equation.
- Result: effective dynamics with **diagonal** multiplicative noise in X and Y. No simultaneous reactions remain, hence no off-diagonal noise.
- A negative feedback $Y \to X$ can be included, e.g. via $Y + R \rightleftharpoons C_R$.

From Chemistry to the Effective Model

Fluctuations around the deterministic fixed point: $\eta_i = H_i - H_i^{(0)}$, $x = X - X^{(0)}$, $y = Y - Y^{(0)}$. Linear Noise Approximation, plus a phenomenological non-linear term $g(x)$ for the finite dynamic range of the sensor:

$$\dot{\eta} = -\tau_\eta^{-1} W_\eta \eta + \sqrt{2D_\eta}\,\xi_\eta$$

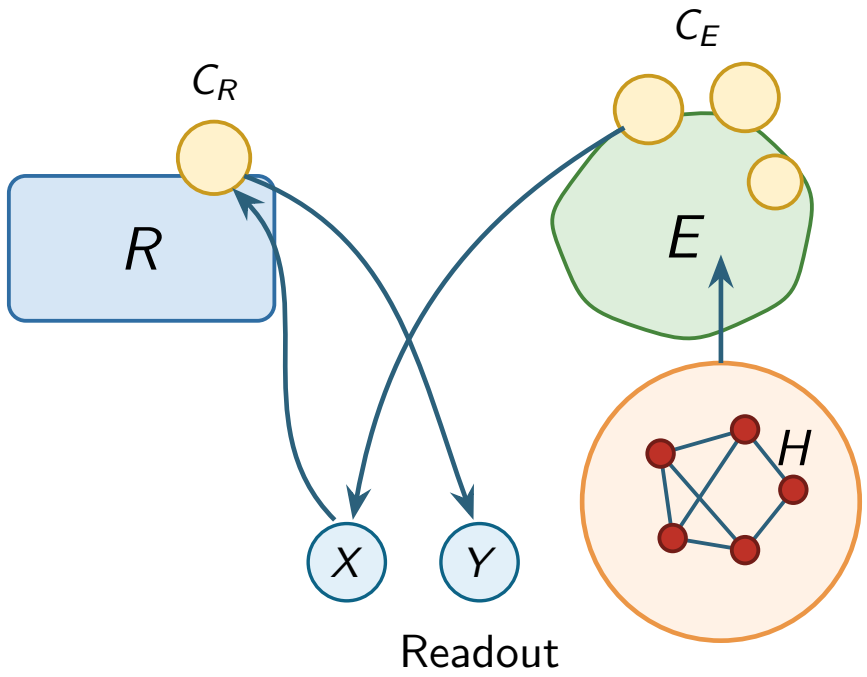
$$\dot{x} = -\tau_x^{-1} x + \sigma \cdot \eta + \alpha a y + \sqrt{2D_x}\,\xi_x$$

$$\dot{y} = -\tau_y^{-1} y + a\, g(x) + \sqrt{2D_y}\,\xi_y$$

- a: tunable coupling; $\alpha < 0$: possible inhibitory feedback; W_η: internal interactions of the hidden degrees of freedom.
- Hierarchical, non-reciprocal structure: $\eta \to x \to y$, no back-action unless $\alpha \neq 0$.
- In this work: $g(x) = x$ and $\alpha = 0$ (strictly feed-forward, linear).

Variable	**Chemical picture**	**Physical picture**
η (active bath)	hidden network H_i	cytoskeleton activity
x (membrane)	signaling molecule X	lipid bilayer
y (probe)	internal readout Y	external sensor (e.g. optical tweezers)

Chemical Transduction Cascade



The hidden network H drives the intermediate complexes C_E, which synthesize the signaling molecule X. X forms the complex C_R upon binding the receptor, ultimately triggering the production of the readout Y.

Reservoir Entropy Rate

- The system's dynamics are governed by the generalized overdamped Langevin equation:

$$\dot{\mathbf{z}}(t) = \boldsymbol{\gamma}^{-1}\mathbf{F}(\mathbf{z}, t) + \sqrt{2\mathbf{D}}\boldsymbol{\xi}(t)$$

- The total average entropy production rate is the sum of the system and reservoir contributions:

$$\langle \dot{s}_{tot}(t) \rangle = \langle \dot{s}_{sys}(t) \rangle + \langle \dot{s}_{res}(t) \rangle$$

- In a NESS, the macroscopic system entropy is constant ($\langle \dot{s}_{sys} \rangle = 0$). By adopting natural dimensionless units ($k_B T = 1$), the $1/T$ factor is absorbed.

- According to stochastic energetics, the resulting reservoir entropy rate (heat dissipation) is mathematically evaluated using the Stratonovich product ($\circ$):

$$\langle \dot{s}_{tot} \rangle = \langle \dot{s}_{res} \rangle = \langle \mathbf{F}(\mathbf{z}) \circ \dot{\mathbf{z}}(t) \rangle$$

Relative Entropy Rate (2)

- **Stochastic Energetics:** Average heat dissipation in a NESS:

$$\dot{S} = \int \boldsymbol{F}(\boldsymbol{z}) \cdot \boldsymbol{J}(\boldsymbol{z})\, d\boldsymbol{z}$$

- Substituting the deterministic force $\boldsymbol{F}(\boldsymbol{z}) = -\boldsymbol{D}^{-1}\boldsymbol{A}\boldsymbol{z}$ and the probability current $\boldsymbol{J}(\boldsymbol{z}) = (\boldsymbol{D}\boldsymbol{\Sigma}^{-1} - \boldsymbol{A})\boldsymbol{z}p(\boldsymbol{z})$:

$$\dot{S} = \mathrm{Tr}[\boldsymbol{A}^T\boldsymbol{D}^{-1}\boldsymbol{A}\boldsymbol{\Sigma}] - \mathrm{Tr}[\boldsymbol{A}]$$

- **Relative Cost:** Isolating the cost of the active sensory coupling from the baseline environmental dissipation:

$$\dot{S}^{rel}(a) = \dot{S}_a - \dot{S}_{a=0} = \mathrm{Tr}[\boldsymbol{A}_a^T\boldsymbol{D}^{-1}\boldsymbol{A}_a\boldsymbol{\Sigma}_a] - \mathrm{Tr}[\boldsymbol{A}_0^T\boldsymbol{D}^{-1}\boldsymbol{A}_0\boldsymbol{\Sigma}_0]$$

Simplifying the Cost: Traffic (1)

- At stationarity holds $\dot{S} = 4\mathcal{T} + \mathcal{G}$
  - **Inflow Rate ($\mathcal{G}$):** phase-space contraction induced by the environment.
  - **Traffic ($\mathcal{T}$):** time-symmetric dynamical activity.
- **Variance-Sum Rule (VSR)** allows estimating the traffic directly from empirical trajectories via stationary cross-correlations:

$$\mathcal{T} = -\frac{1}{4}\sum_{i,j}(\boldsymbol{D}^{-1})_{ij}\ddot{C}_{ij}^{(z)}(0)$$

Since our $\boldsymbol{D}$ is diagonal:

$$4\mathcal{T} = \mathrm{Tr}[\boldsymbol{D}^{-1}\boldsymbol{A}\boldsymbol{\Sigma}\boldsymbol{A}^{T} - 2\boldsymbol{A}]$$

Multiple state model: Equations



$$
\begin{cases}
\dot{y} = -\frac{y}{\tau_y} + \frac{1}{\tau_y}\sum_{i=1}^{M} a_i x_i + \sqrt{\frac{2D_y}{\tau_y}}\xi_y(t) & \\
\dot{x}_i = -\frac{x_i}{\tau_{x_i}} + \sum_{\substack{l=1 \\ l\neq i}}^{M} \frac{k_{il}^{(x)}}{\tau_{x_i}} x_l + \sum_{j=1}^{N} \frac{\sigma_{ij}}{\tau_{x_i}}\eta_j + \sqrt{\frac{2D_{x_i}}{\tau_{x_i}}}\xi_{x_i}(t) & i = 1, \ldots, M \\
\dot{\eta}_j = -\frac{\eta_j}{\tau_{\eta_j}} + \sum_{\substack{l=1 \\ l\neq j}}^{N} \frac{k_{jl}^{(\eta)}}{\tau_{\eta_j}} \eta_l + \sqrt{\frac{2D_{\eta_j}}{\tau_{\eta_j}}}\xi_{\eta_j}(t) & j = 1, \ldots, N
\end{cases}
$$

Evaluating the PID Components

For a Gaussian stationary state, the PID components are analytically derived from the covariance matrix $\boldsymbol{\Sigma}$ in four sequential steps:

1. **Mutual Information components:**

$$I(x_i; y) = \frac{1}{2} \ln \left[\frac{\Sigma_{x_i x_i} \Sigma_{yy}}{\Sigma_{x_i x_i} \Sigma_{yy} - \Sigma_{x_i y}^2} \right], \quad I(x_1, x_2; y) = \frac{1}{2} \ln \left[\frac{\Sigma_{yy} \det(\boldsymbol{\Sigma}_{xx})}{\det(\boldsymbol{\Sigma}_{joint})} \right]$$

2. **Redundancy (MMI):**

$$R(x_1, x_2; y) = \min\{I(x_1; y), I(x_2; y)\}$$

3. **Unique Information:** Marginal information minus the redundant overlap.

$$U(x_1 \setminus x_2; y) = I(x_1; y) - R, \quad U(x_2 \setminus x_1; y) = I(x_2; y) - R$$

4. **Synergy:** Extracted via the global information balance.

$$S(x_1, x_2; y) = I(x_1, x_2; y) - I(x_1; y) - I(x_2; y) + R$$

Equal-Dissipation Constraint

Architectures often perform better simply by dissipating more energy. We ensure fair comparisons by fixing the thermodynamic budget:

Equal-Dissipation Constraint

$$\dot{S}^{rel}_{M,N}(a_1, \ldots, a_M) = \dot{S}^{rel}_{M=1,N=1}(a^*) \implies f(a_1, \ldots, a_M) = C$$

Dimensional Reduction:

- Isolate one parameter analytically, e.g., $a_1 = g(a_2, \ldots, a_M)$.
- Restrict the search to an $(M-1)$-dimensional manifold preserving total energy.
- The optimization becomes a maximization over the reduced space:

$$\mathbf{a}^* = \underset{a_2,\ldots,a_M}{\operatorname{argmax}} \mathcal{L}\big(g(a_2, \ldots, a_M), a_2, \ldots, a_M\big)$$

Topology $M = 2$, $N = 1$: Matrices, Stability, Constraint



Vectorial Langevin dynamics: $\dot{\mathbf{z}} = -A\mathbf{z} + \tilde{D}\boldsymbol{\xi}(t)$

$$A = \begin{pmatrix} \frac{1}{\tau_y} & -\frac{a_1}{\tau_y} & -\frac{a_2}{\tau_y} & 0 \\ 0 & \frac{1}{\tau_{x_1}} & -\frac{k}{\tau_{x_1}} & -\frac{\sigma}{\tau_{x_1}} \\ 0 & -\frac{k}{\tau_{x_2}} & \frac{1}{\tau_{x_2}} & -\frac{\sigma}{\tau_{x_2}} \\ 0 & 0 & 0 & \frac{1}{\tau_\eta} \end{pmatrix} \qquad \tilde{D} = \sqrt{2}\begin{pmatrix} \sqrt{\frac{D_y}{\tau_y}} & 0 & 0 & 0 \\ 0 & \sqrt{\frac{D_{x_1}}{\tau_{x_1}}} & 0 & 0 \\ 0 & 0 & \sqrt{\frac{D_{x_2}}{\tau_{x_2}}} & 0 \\ 0 & 0 & 0 & \sqrt{\frac{D_\eta}{\tau_\eta}} \end{pmatrix}$$

1. Stability Conditions

Block-triangular structure $\Longrightarrow$ stability is dictated solely by the membrane layer A_{xx}:

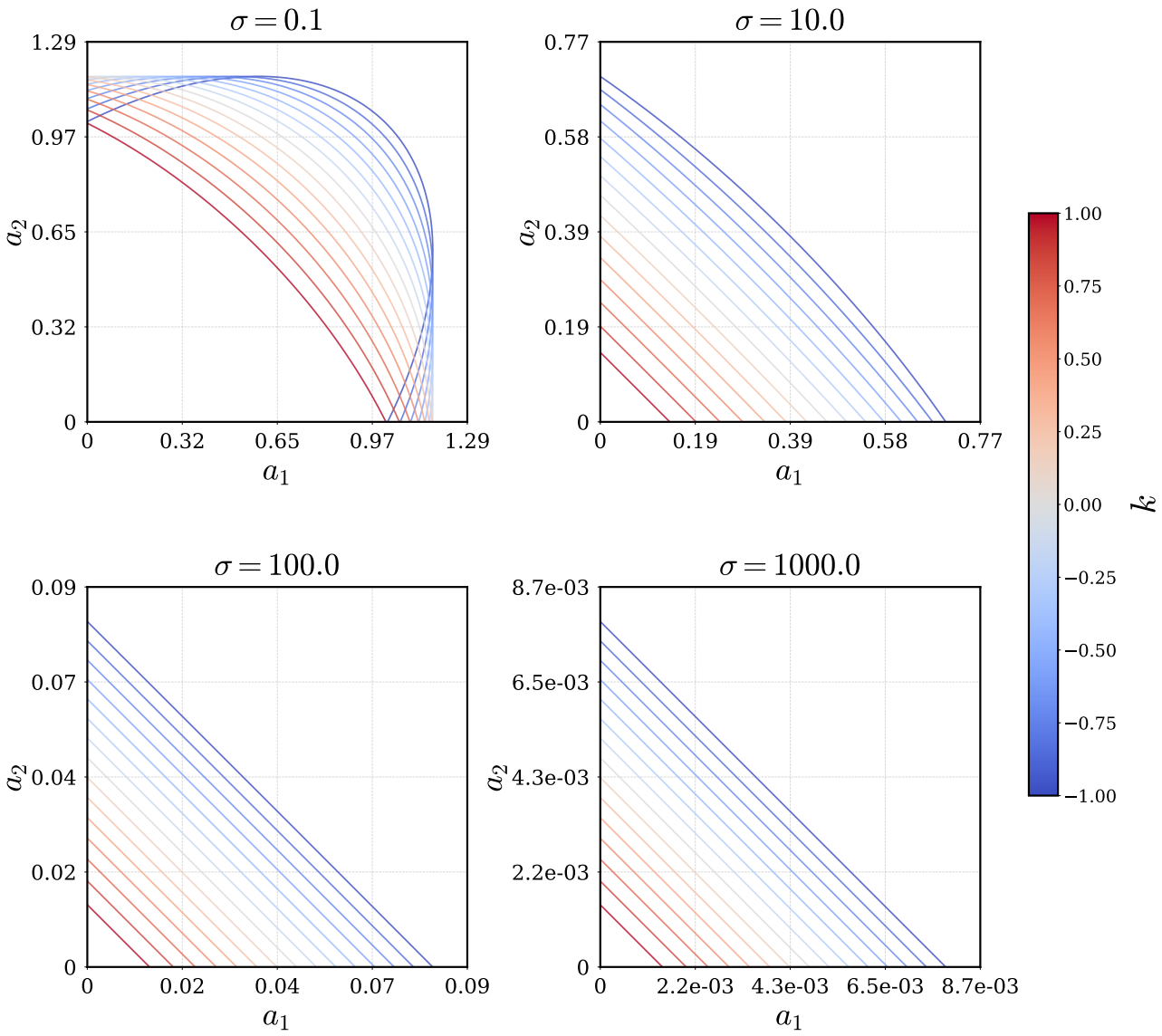
$$\lambda_\pm = \frac{1 \pm k}{\tau_x} \qquad \Longrightarrow \qquad \text{Stable iff } |\mathbf{k}| < \mathbf{1}$$

2. Equal-Dissipation Constraint

The parameter search is restricted to an **elliptical manifold** ($\mathcal{B}^2 - 4\mathcal{A}\mathcal{C} < 0$) for $a_1, a_2 \geq 0$:

$$\mathcal{A}(k,\sigma,\tau)a_1^2 + \mathcal{B}(k,\sigma,\tau)a_1 a_2 + \mathcal{C}(k,\sigma,\tau)a_2^2 = \dot{S}^{rel}_{M=1,N=1}(a^*)\mathcal{Q}(k,\tau)$$

Equal Dissipation plot



Topology $M = 1$, $N = 2$: Matrices, Stability, Constraint

Vectorial Langevin dynamics: $\dot{\mathbf{z}} = -A\mathbf{z} + \tilde{D}\boldsymbol{\xi}(t)$

$$A = \begin{pmatrix} \frac{1}{\tau_y} & -\frac{a}{\tau_y} & 0 & 0 \\ 0 & \frac{1}{\tau_x} & -\frac{\sigma_1}{\tau_x} & -\frac{\sigma_2}{\tau_x} \\ 0 & 0 & \frac{1}{\tau_\eta} & -\frac{k}{\tau_\eta} \\ 0 & 0 & -\frac{k}{\tau_\eta} & \frac{1}{\tau_\eta} \end{pmatrix} \qquad \tilde{D} = \sqrt{2}\begin{pmatrix} \sqrt{\frac{D_y}{\tau_y}} & 0 & 0 & 0 \\ 0 & \sqrt{\frac{D_x}{\tau_x}} & 0 & 0 \\ 0 & 0 & \sqrt{\frac{D_{\eta_1}}{\tau_\eta}} & 0 \\ 0 & 0 & 0 & \sqrt{\frac{D_{\eta_2}}{\tau_\eta}} \end{pmatrix}$$

1. Stability Conditions

Block-triangular structure $\Longrightarrow$ stability is dictated solely by the active baths layer $A_{\eta\eta}$:

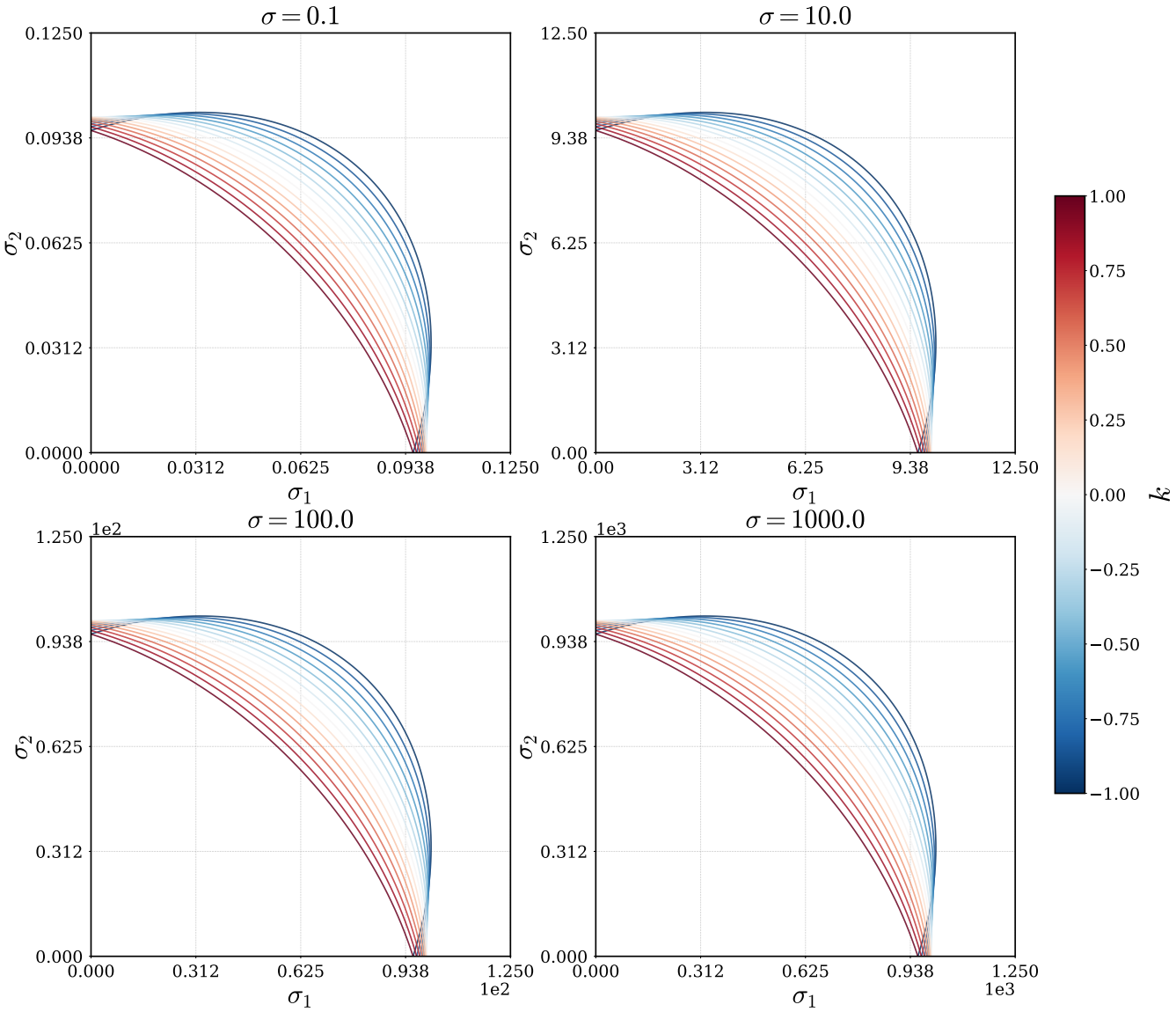
$$\lambda_\pm = \frac{1 \pm k}{\tau_\eta} \qquad \Longrightarrow \qquad \text{Stable iff } |\mathbf{k}| < \mathbf{1}$$

2. Equal-Dissipation Constraint

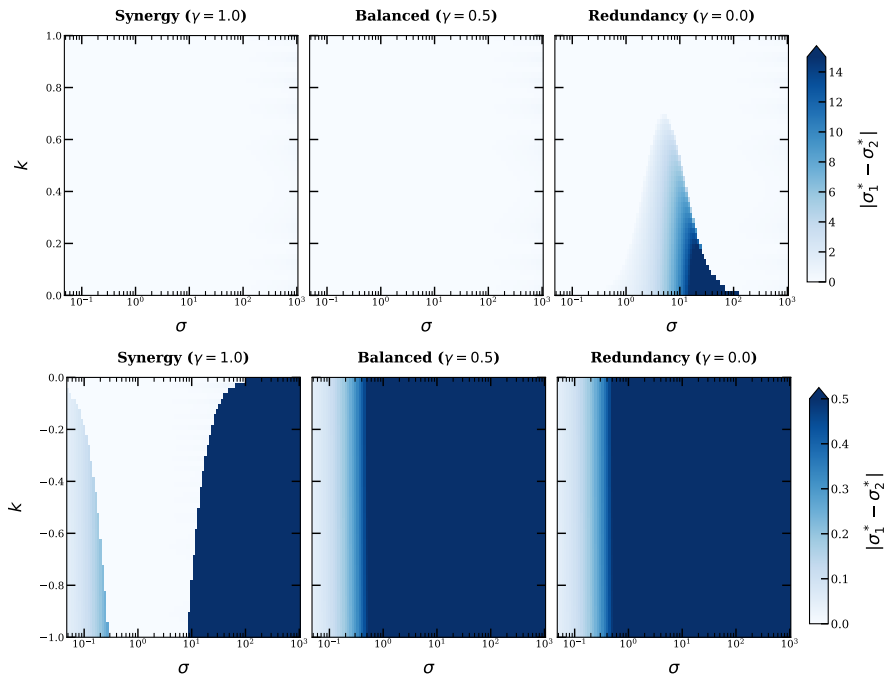
The parameter search is restricted to an **elliptical manifold** ($\mathcal{B}^2 - 4\mathcal{A}\mathcal{C} < 0$) for $\sigma_1, \sigma_2 \geq 0$. Notably, the sensor coupling $(a^*)^2$ cancels out:

$$\mathcal{A}(k,\tau)\sigma_1^2 + \mathcal{B}(k,\tau)\sigma_1\sigma_2 + \mathcal{C}(k,\tau)\sigma_2^2 = \tilde{S}^{rel}_{M=1,N=1}(\sigma,\tau)\delta(k,\tau) - \Phi(k,\tau)$$

Equal Dissipation Costraint



Coupling Asymmetries $|\sigma_1^* - \sigma_2^*|$



- $k < 0 \rightarrow$ to avoid destructive interference, in Balanced and Redundancy-driven the system immediately shuts down one sensory coupling, securing a macroscopic information gain ($\Delta I > 0$).
- $k < 0 \rightarrow$ Synergy maximization forces alternating asymmetric/symmetric regions.

Total Entropy in a NESS

- The system's dynamics are governed by the generalized overdamped Langevin equation:

$$\dot{\mathbf{z}}(t) = \boldsymbol{\gamma}^{-1}\mathbf{F}(\mathbf{z}, t) + \sqrt{2\mathbf{D}}\boldsymbol{\xi}(t)$$

- In a Non-Equilibrium Steady State (NESS), the total average entropy production rate can be exactly decomposed into two distinct contributions:

$$\langle \dot{s}_{tot} \rangle = 4\mathcal{T} + \mathcal{G}$$

- **Traffic ($\mathcal{T}$):** A time-symmetric kinetic observable characterizing the dynamical activity (or *frenesy*) of the system, insensitive to time-reversal.
- **Inflow Rate ($\mathcal{G}$):** The continuous-state equivalent of the phase-space contraction rate induced by the environment.

The Mean Local Velocity

- Assuming autonomous fields $\mathbf{F}(\mathbf{z})$ and a diagonal diffusion tensor $\mathbf{D}$ (uncorrelated white noise), we define the **mean local velocity**:

$$\boldsymbol{\nu}(\mathbf{z}) = \frac{\mathbf{J}(\mathbf{z})}{p(\mathbf{z})} = \mathbf{D}\left(\mathbf{F}(\mathbf{z}) - \nabla_{\mathbf{z}} \log p(\mathbf{z})\right)$$

- **Physical Interpretation:** The physical force field splits into a part directly tied to the observable mean drift ($\boldsymbol{\nu}$), and a part tied to the local curvature of the stationary distribution.

The Kinetic Quadratic Form

- The total entropy production is defined as $\dot{S} = \langle \mathbf{F}(\mathbf{z})^T \boldsymbol{\nu}(\mathbf{z}) \rangle$.
- Substituting $\mathbf{F}(\mathbf{z})$ and applying the NESS condition, the cross-term linking the density gradient to the mean velocity vanishes. Entropy production becomes purely kinetic and non-negative:

$$\dot{S} = \langle \boldsymbol{\nu}(\mathbf{z})^T \mathbf{D}^{-1} \boldsymbol{\nu}(\mathbf{z}) \rangle$$

- Expanding this quadratic form yields three components:
  1. A pure force term.
  2. A mixed cross-term (force and probability gradient).
  3. A purely geometric term tied to $\nabla_{\mathbf{z}} \log p(\mathbf{z})$.

Resolving the Cross-Term

- The direct expansion of the kinetic form is:

$$\dot{S} = \langle \mathbf{F}^T \mathbf{D} \mathbf{F} \rangle - 2 \langle \mathbf{F}^T \mathbf{D} \nabla \log p \rangle + \langle (\nabla \log p)^T \mathbf{D} \nabla \log p \rangle$$

- We evaluate the mixed cross-term separately using integration by parts, assuming the probability density vanishes at infinity:

$$\langle \mathbf{F}(\mathbf{z})^T \mathbf{D} \nabla_{\mathbf{z}} \log p(\mathbf{z}) \rangle = -\langle \nabla_{\mathbf{z}} \cdot (\mathbf{D}\mathbf{F}(\mathbf{z})) \rangle$$

- This converts the mixed term into the expected divergence of the deterministic drift vector.

The Stationary Relation

- In a NESS, the probability current is divergence-free ($\nabla_{\mathbf{z}} \cdot \mathbf{J}(\mathbf{z}) = 0$). This implies that the expected divergence of the mean local velocity is zero ($\langle \nabla_{\mathbf{z}} \cdot \boldsymbol{\nu}(\mathbf{z}) \rangle = 0$).
- Applying integration by parts yields the fundamental stationary-state relation:

$$\langle (\nabla_{\mathbf{z}} \log p(\mathbf{z}))^T \mathbf{D} \nabla_{\mathbf{z}} \log p(\mathbf{z}) \rangle = -\langle \nabla_{\mathbf{z}} \cdot (\mathbf{DF}(\mathbf{z})) \rangle$$

- **Implication:** The purely geometric term exactly matches (up to a sign) the divergence of the drift, enabling the removal of explicit dependencies on $p(\mathbf{z})$.

Traffic ($4\mathcal{T}$) and Inflow Rate ($\mathcal{G}$)

- Applying the stationary relation, we can finally regroup the expression into our two macroscopic observables:

- **Traffic ($4\mathcal{T}$):** Depends purely on the physical force field, measuring time-symmetric dynamical activity.

$$4\mathcal{T} = \langle \mathbf{F}(\mathbf{z})^{T} \mathbf{DF}(\mathbf{z}) \rangle + 2\langle \nabla_{\mathbf{z}} \cdot (\mathbf{DF}(\mathbf{z})) \rangle$$

- **Inflow Rate ($\mathcal{G}$):** Quantifies probability compression/expansion driven by the deterministic drift.

$$\mathcal{G} = -\langle \nabla_{\mathbf{z}} \cdot (\mathbf{DF}(\mathbf{z})) \rangle$$

Mutual Information

- **Stationary Fokker-Planck** ($\boldsymbol{D} = \frac{1}{2}\tilde{\boldsymbol{D}}\tilde{\boldsymbol{D}}^T$):

$$\sum_{i,j} \frac{\partial}{\partial z_i}(A_{ij} z_j p(\boldsymbol{z})) + \sum_{i,j} D_{ij} \frac{\partial^2}{\partial z_i \partial z_j} p(\boldsymbol{z}) = 0$$

  $\rightarrow$ Multivariate Gaussian distribution $p(\boldsymbol{z})$.

- **Lyapunov Equation:** $\boldsymbol{A\Sigma} + \boldsymbol{\Sigma A}^T = 2\boldsymbol{D}$
- Differential entropies $h(z) = -\int p(\boldsymbol{z}) \ln p(\boldsymbol{z})\, d\boldsymbol{z}$ computed for x, y, and (x, y)

$$\begin{aligned} I(x; y) &= h(x) + h(y) - h(x, y) \\ &= \frac{1}{2} \ln \left[\frac{\Sigma_{xx}\Sigma_{yy}}{\Sigma_{xx}\Sigma_{yy} - \Sigma_{xy}^2} \right] \end{aligned}$$

Mutual Information in Gaussian States

- Using the chain rule for differential entropy, the Mutual Information between any two variables (e.g., membrane x and probe y) is:

$$I(x;y) = h(x) + h(y) - h(x,y)$$

- Since the unforced system relaxes to a NESS with a zero-mean multivariate Gaussian distribution, we can compute these quantities analytically.

- The marginal differential entropy for a 1D variable x with variance Σ_{xx} evaluates exactly to:

$$h(x) = -\int_{-\infty}^{+\infty} p(x) \ln p(x)\, dx = \frac{1}{2} \ln(2\pi e \Sigma_{xx})$$

Bivariate Joint Entropy



- For the bivariate state vector $\mathbf{w} = (x, y)^T$, the joint entropy is $h(x, y) = -\mathbb{E}[\ln p(\mathbf{w})]$. Substituting the Gaussian density gives:

$$h(x, y) = \ln(2\pi) + \frac{1}{2}\ln(\det(\boldsymbol{\Sigma}_{xy})) + \frac{1}{2}\mathbb{E}\left[\mathbf{w}^T\boldsymbol{\Sigma}_{xy}^{-1}\mathbf{w}\right]$$

- To evaluate the expected value of the quadratic form, we exploit the cyclical property of the trace ($\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$):

$$\mathbb{E}\left[\mathbf{w}^T\boldsymbol{\Sigma}_{xy}^{-1}\mathbf{w}\right] = \mathrm{Tr}\left(\boldsymbol{\Sigma}_{xy}^{-1}\mathbb{E}\left[\mathbf{w}\mathbf{w}^T\right]\right)$$

- Since $\mathbb{E}[\mathbf{w}\mathbf{w}^T] = \boldsymbol{\Sigma}_{xy}$, this term reduces to $\mathrm{Tr}(\mathbf{I}_{2\times 2}) = 2$, yielding:

$$h(x, y) = \frac{1}{2}\ln\left((2\pi e)^2\det(\boldsymbol{\Sigma}_{xy})\right)$$

Scalar Mutual Information Formula



- Substituting the marginal and joint entropies back into the general mutual information equation:

$$I(x;y) = \frac{1}{2}\left[\ln(\Sigma_{xx}) + \ln(\Sigma_{yy}) - \ln(\det(\boldsymbol{\Sigma}_{xy}))\right]$$

- Expanding the determinant of the 2×2 covariance submatrix ($\det(\boldsymbol{\Sigma}_{xy}) = \Sigma_{xx}\Sigma_{yy} - \Sigma_{xy}^2$), we obtain the exact analytical expression:

$$I(x;y) = \frac{1}{2}\ln\left[\frac{\Sigma_{xx}\Sigma_{yy}}{\Sigma_{xx}\Sigma_{yy} - \Sigma_{xy}^2}\right]$$

- This explicit formula characterizes the information extraction limit for the single-membrane architecture ($M = 1, N = 1$).

Vectorial Generalization

- In generalized topologies, the source variable is a d-dimensional vector $\mathbf{x} \in \mathbb{R}^d$ (representing M membranes or N baths), while the probe y remains scalar.
- The joint covariance matrix of the $(d+1)$-dimensional coupled system is structured in block form:

$$\boldsymbol{\Sigma}_{\text{joint}} = \begin{pmatrix} \Sigma_{yy} & \boldsymbol{\Sigma}_{y\mathbf{x}} \\ \boldsymbol{\Sigma}_{\mathbf{x}y} & \boldsymbol{\Sigma}_{\mathbf{xx}} \end{pmatrix}$$

- Applying the same entropy chain rule to the multivariate distributions yields the generalized block-determinant expression:

$$I(\mathbf{x}; y) = \frac{1}{2} \ln \left(\frac{\Sigma_{yy} \det(\boldsymbol{\Sigma}_{\mathbf{xx}})}{\det(\boldsymbol{\Sigma}_{\text{joint}})} \right)$$

Exponential scaling of PID

